

Multi-species Catalogues of Regulatory Elements: Ensembl

Session 3

T6: Making the most of cis-regulatory annotations for genome-scale analysis

Louisse Paola Mirabueno - Garth Ilesley

Ensembl Outreach Officer,
Ensembl Outreach,
EMBL-EBI

Project Lead
Ensembl Regulation,
EMBL-EBI

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Session map

The Ensembl Project	• General description
Ensembl Regulation	• Regulatory pipelines • Available datasets and Regulatory annotation • Defining and accessing reference epigenome data
The Ensembl Regulation subsite	• Epigenomes data and metadata • The regulatory features tab • Exploring genome annotation and regulatory activity
Comparison of CREs frameworks	• Most-popular genome-wide resources • Strengths and weaknesses of each approach
Hands-on	

The Ensembl Project

The mission of Ensembl is to provide **comprehensive and accurate annotation** of genomes, **integrating** this information with other biological data and making it **publicly accessible** via the web.



ENSEMBL

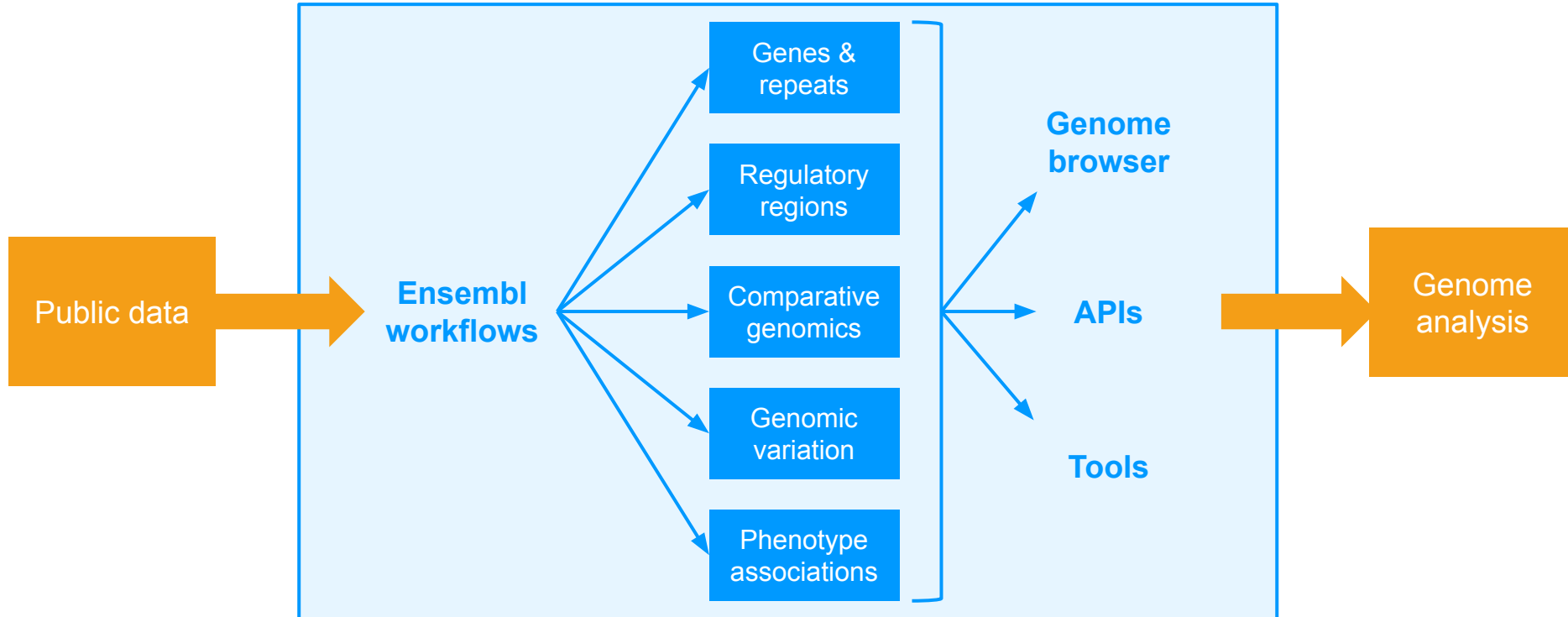
Using genome browsers to visualise annotation

Regulatory element

Gene

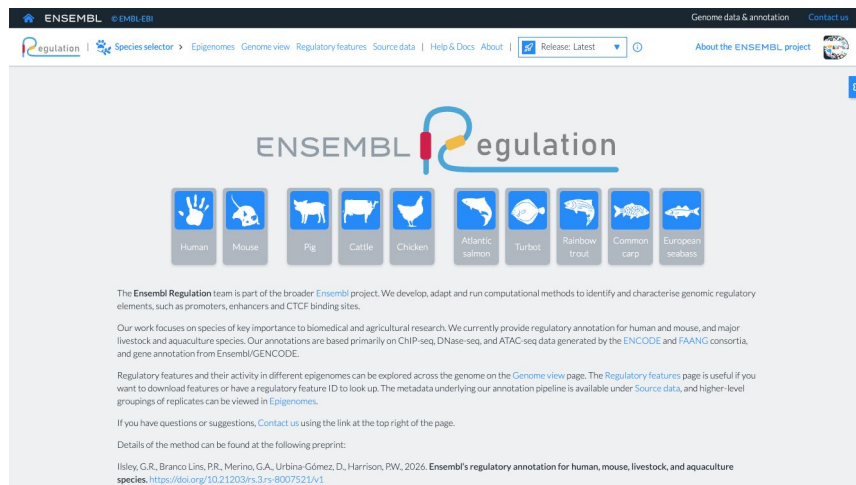
Variant

Ensembl enables genome analyses



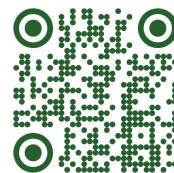
Ensembl Regulation

- Aim: Develop, adapt and run computational methods to identify and characterise genomic regulatory elements, such as promoters, enhancers and CTCF binding sites.
- Focus: Species of key importance to biomedical and agricultural research.
- Annotations are based primarily on ChIP-seq, DNase-seq, and ATAC-seq data generated by the ENCODE and FAANG consortia, and gene annotation from Ensembl/GENCODE.

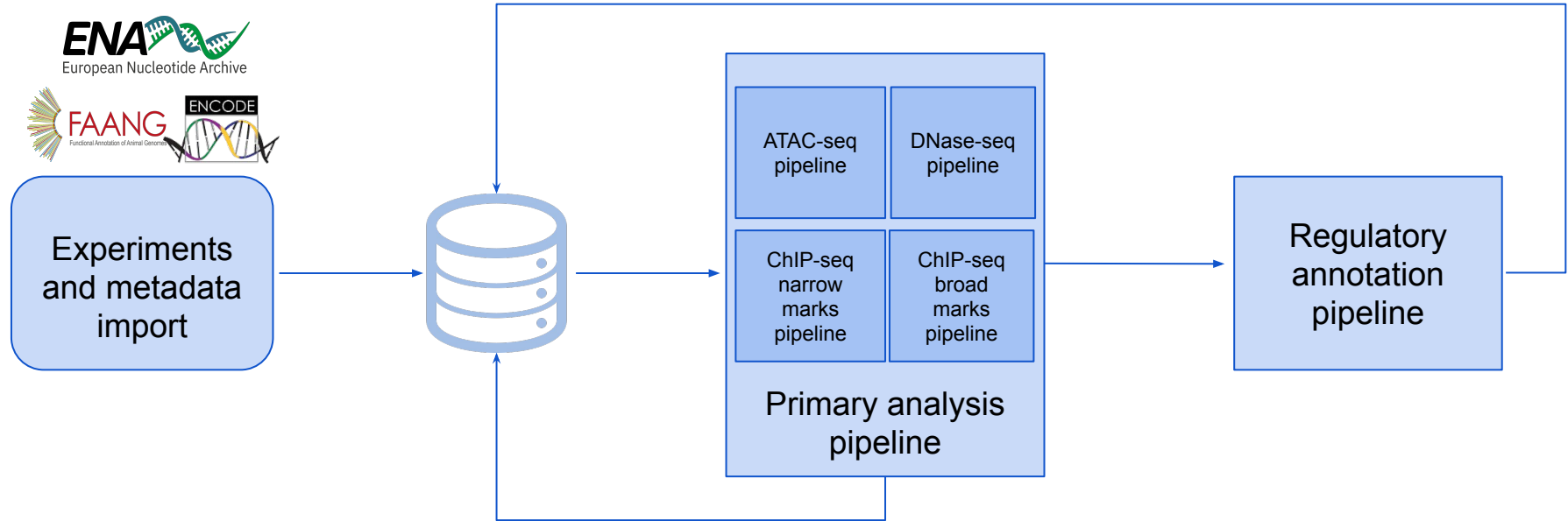


The screenshot shows the Ensembl Regulation website. At the top is a navigation bar with the Ensembl logo, a search bar, and links for 'Genome data & annotation' and 'Contact us'. Below this is a secondary navigation bar with links for 'Regulation', 'Species selector', 'Epigenomes', 'Genome view', 'Regulatory features', 'Source data', 'Help & Docs', 'About', and a 'Release: Latest' dropdown. The main content area features the 'ENSEMBL Regulation' logo, which includes a stylized DNA double helix. Below the logo is a row of icons representing different species: Human, Mouse, Pig, Cattle, Chicken, Atlantic salmon, Turbot, Rainbow trout, Common carp, and European seabass. Each icon is a blue square with a white silhouette of the species. Below the species icons, there is a paragraph of text: 'The Ensembl Regulation team is part of the broader Ensembl project. We develop, adapt and run computational methods to identify and characterise genomic regulatory elements, such as promoters, enhancers and CTCF binding sites. Our work focuses on species of key importance to biomedical and agricultural research. We currently provide regulatory annotation for human and mouse, and major livestock and aquaculture species. Our annotations are based primarily on ChIP-seq, DNase-seq, and ATAC-seq data generated by the ENCODE and FAANG consortia, and gene annotation from Ensembl/GENCODE. Regulatory features and their activity in different epigenomes can be explored across the genome on the Genome view page. The Regulatory features page is useful if you want to download features or have a regulatory feature ID to look up. The metadata underlying our annotation pipeline is available under Source data, and higher-level groupings of replicates can be viewed in Epigenomes. If you have questions or suggestions, Contact us using the link at the top right of the page. Details of the method can be found at the following preprint: Ilisley, G.R., Branco Lins, P.R., Merino, G.A., Urbina-Gómez, D., Harrison, P.W., 2026. Ensembl's regulatory annotation for human, mouse, livestock, and aquaculture species. <https://doi.org/10.21203/rs.3.rs-8007521/v1>

<https://regulation.ensembl.org/>



The Ensembl Regulatory Build

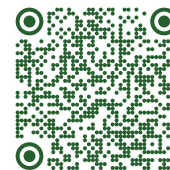


Primary analysis pipeline

- Processes read files to produce peak (BED) and signal (bigWig) files.



- Preprocessing: Adapters trimming with NGmerge/fastp.
- Alignment: Bowtie2.
- Peak calling:
 - Genome mask:
 - ENCODE masks for human and mouse.
 - Custom masking strategy for other species.
 - Genrich with different configurations for each experiment type.
- Signal: Estimated from Genrich pileups



Standardised data analysis pipelines

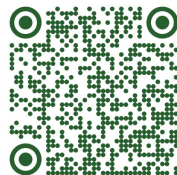
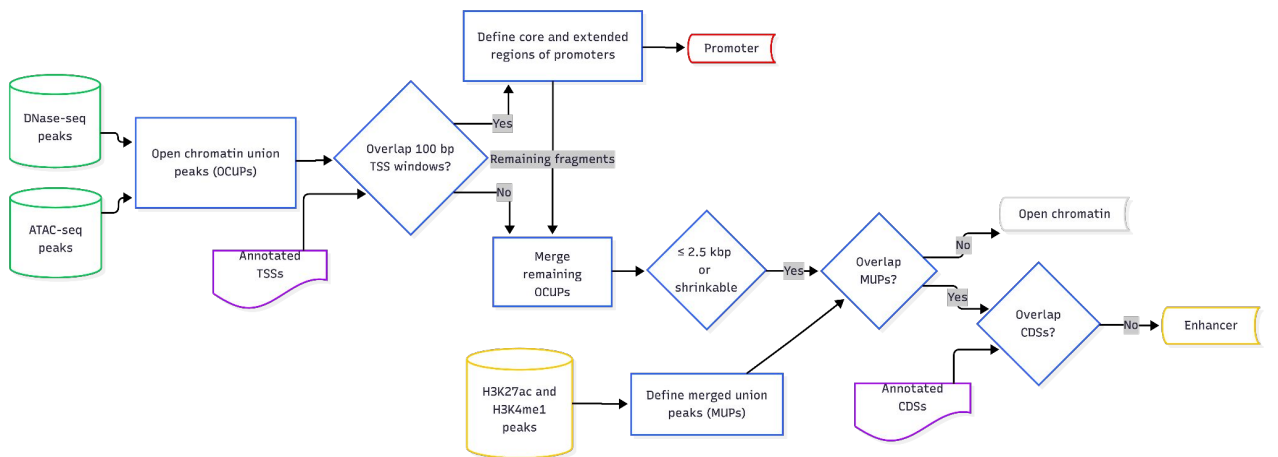
Regulatory annotation pipeline

Input:

- **ATAC-seq & DNase-seq** peaks for all the epigenomes of the target species
- H3K27ac, H3K4me1, and H3K4me3 ChIP-seq peaks for all the epigenomes of the target species (**optional**)
- Annotated TSSs and CDSs from transcripts of interest

Output:

- Set of regulatory features: **Promoters, Enhancers, OCR**
- Regulatory features (RFs) **activity** across different epigenomes with open **chromatin** data



The Ensembl Regulation Subsite

Contact form

Page selector

Genome data & annotation

Contact us

Regulation



Species selector

Epigenomes

Genome view

Regulatory features

Source data

Help & Docs

About



Release: Latest

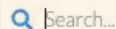


About the ENSEMBL project



Release

Configuration tool



Search...



Human (*Homo sapiens*)



Mouse (*Mus musculus*)



Pig (*Sus scrofa*)



Cattle (*Bos taurus*)



Chicken (*Gallus gallus*)



Atlantic salmon (*Salmo salar*)



Turbot (*Scophthalmus maximus*)



Rainbow trout (*Oncorhynchus mykiss*)



Common carp (*Cyprinus carpio carpio*)

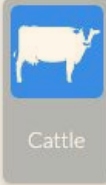


European seabass (*Dicentrarchus labrax*)

ENSEMBL Regulation



Pig



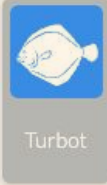
Cattle



Chicken



Atlantic salmon



Turbot



Rainbow trout



Common carp



European seabass

Species selector

in computational methods to identify and characterise genomic regulatory

Our work focuses on species of key importance to biomedical and agricultural research. We currently provide regulatory annotation for human and mouse, and major livestock and aquaculture species. Our annotations are based primarily on ChIP-seq, DNase-seq, and ATAC-seq data generated by the [ENCODE](#) and [FAANG](#) consortia, and gene annotation from Ensembl/GENCODE.

Epigenomes tab

GRCh38

Showing 144 epigenomes.

Epigenome	Biosample term	Tissue group	Sex	Life stage	Age	Biosample material	Assays
A549 (m, 58 y)	A549	lung	male	adult	58 y	cell line	
A673 (f, 15 y)	A673	muscle	female	child	15 y	cell line	
ACC112 (m, 70 y)	ACC112	epithelium	male	adult	70 y	cell line	
adrenal gland (f, 53 y)	adrenal gland	endocrine gland	female	adult	53 y	primary tissue	
adrenal gland (m, 37 y)	adrenal gland	endocrine gland	male	adult	37 y	primary tissue	
adrenal gland (m, 54 y)	adrenal gland	endocrine gland	male	adult	54 y	primary tissue	
adrenal gland (96-97 dpf)	adrenal gland	endocrine gland	male, unknown	embryonic	96-97 dpf	primary tissue	

GRCh38 / A549 (m, 58 y)

ATAC-seq 3 biological replicates						
Source	Biosample	Biomaterial type	Term	Life stage	Sex	Age
ENCSR032RGS	ENCB05024PJA	cell line	A549	adult	male	58 years
ENCSR032RGS	ENCB0503HWJ	cell line	A549	adult	male	58 years
ENCSR032RGS	ENCB05049MEE	cell line	A549	adult	male	58 years

DNase-seq 2 biological replicates						
Source	Biosample	Biomaterial type	Term	Life stage	Sex	Age
ENCSR000ELW	ENCB05163AAA	cell line	A549	adult	male	58 years
ENCSR000ELW	ENCB05858LUL	cell line	A549	adult	male	58 years

ChIP-seq (CTCF) 1 biological replicate						
Source	Biosample	Biomaterial type	Term	Life stage	Sex	Age
ENCSR000DNA	ENCB05162AAA	cell line	A549	adult	male	58 years

ChIP-seq (H3K4me3) 2 biological replicates						
Source	Biosample	Biomaterial type	Term	Life stage	Sex	Age
ENCSR000DPD	ENCB05858LUL	cell line	A549	adult	male	58 years
ENCSR000DPD	ENCB05163AAA	cell line	A549	adult	male	58 years

Epigenomes tab



GRCh38

Showing 4 epigenomes.

Epigenome	Biosample term	Tissue group	Sex	Life stage	Age	Biosample material	Assays
astrocyte	astrocyte			unknown		primary cell	
brain (120-122 dpf)	brain			embryonic	120-122 dpf	primary tissue	
neural progenitor cell (f, 5 dpf)	neural progenitor cell			embryonic	5 dpf	primary cell culture	
SK-N-SH (f, 4 y)	SK-N-SH			child	4 y	cell line	

Move

Deselect all - Select all

Search...

- blood ☐
- bone marrow ☐
- brain ☒
- connective tissue ☐
- embryonic ☐
- endocrine gland ☐
- epithelium ☐
- gonad ☐
- heart ☐
- immune organ ☐
- intestine ☐
- kidney ☐
- limb ☐

Regulatory Features tab

GRCh38

Search...

15 entries per page

Showing 1 to 15 of 643528 features

Download

ID	Type	Location	Genes
ENSR00001408561	EMAR	1:9,953-11,649	
ENSR1_958	Promoter	1:10,936-11,436	ENSG00000290825
ENSR1_53X2	CTCF	1:11,222-11,243	
ENSR1_53X9	CTCF	1:11,280-11,301	
ENSR1_53XJ	CTCF	1:11,339-11,360	
ENSR1_88N	Enhancer	1:11,437-11,649	
ENSR1_CZ	Enhancer	1:12,686-14,671	
ENSR00001408565	EMAR	1:12,686-14,671	
ENSR1_B48	Enhancer	1:15,101-15,853	

GRCh38

promoter

15 entries per page

Showing 1 to 15 of 35983 features

Download

ID	Type	Location	Genes
ENSR1_958	Promoter	1:10,936-11,436	ENSG00000290825
ENSR1_9B7	Promoter	1:28,220-28,599	ENSG00000243485
ENSR1_8M3	Promoter	1:29,333-29,564	ENSG00000310526, ENSG00000243485
ENSR1_9BN	Promoter	1:29,777-30,277	ENSG00000243485
ENSR1_73K8	Promoter	1:36,436-36,536	ENSG00000308361
ENSR1_843W	Promoter	1:96,548-96,720	ENSG00000308314
ENSR1_93XG	Promoter	1:181,117-181,517	ENSG00000241860, ENSG00000308415

Regulatory Features tab



GRCh38 / ENSR1_9B7

Click to visualise location

Location **1: 28,220–28,599**

Genes: [ENSG00000243485](#)

Activity

A feature is Active if an open chromatin peak overlaps the feature in that epigenome.

Active (22/112)

- A549 (m, 58 y)
- adrenal gland (96-97 dpf)
- adrenal gland (m, 37 y)
- body of pancreas (m, 37 y)
- DND-41 (m, 13 y)
- GM23248 (m, 53 y)
- GM23338 (m, 53 y)
- heart (91-105 dpf)
- IMR-90 (f, 16 w)
- K562 (f, 53 y)
- Karpas-422 (f, 72 y)
- kidney (f, 120 dpf)
- mesendoderm (m)
- MM.1S (f, 42 y)

Inactive (90/112)

- A673 (f, 15 y)
- adrenal gland (f, 53 y)
- adrenal gland (m, 54 y)
- AG04450 (m, 12 w)
- ascending aorta (f, 51 y)
- ascending aorta (f, 53 y)
- astrocyte
- B cell (f, 27-43 y)
- B cell (m, 21-37 y)
- BJ (m)
- body of pancreas (f, 53 y)
- body of pancreas (m, 54 y)
- brain (120-122 dpf)
- breast epithelium (f, 51 y)

Genome view tab

The screenshot displays the ENSEMBL Genome view interface. At the top, the navigation bar includes links for Regulation, Human, Epigenomes, Genome view, Regulatory features, Source data, Help & Docs, and About, along with a 'Release: Latest' dropdown and a link to 'About the ENSEMBL project'. The main content area shows a genomic track for GRCh38. A search bar at the top right contains the coordinates '1:28,220-28,599' and a 'Go' button. Below the search bar, the 'Current location' is indicated by a box around the coordinates '1:28,220-28,599'. The 'Zoom tools' are represented by minus and plus buttons. The main track displays various genomic features: Genes (with WASH7P and MIR2), Regulatory features (with Promoter, Enhancer, Open chromatin, EMAR, and CTCF binding site), and a 'Regulatory features card' showing 'Regulatory features 2'. Below this, the 'Filtered epigenomes' section shows a list of epigenomes: astrocyte, brain (120-122 dpf), neural progenitor cell (l. 5 dpf), and SK-N-SH (l. 4 y). The bottom of the screen shows a detailed view of the epigenomes, with a legend indicating H3K27ac, H3K4me1, and H3K27me2.

Regulation | Human > Epigenomes | **Genome view** | Regulatory features | Source data | Help & Docs | About | Release: Latest | About the ENSEMBL project

GRCh38

Search coordinates

1:28,220-28,599

Go

1:28,220-28,599

Current location

Zoom tools

Genes

WASH7P

MIR2

Regulatory features

Promoter Enhancer Open chromatin EMAR Epigenetically-modified accessible region CTCF CTCF binding site

Regulatory features card

Regulatory features 2

Regulatory activity 4 epigenomes

Filtered epigenomes

Epigenomes

astrocyte

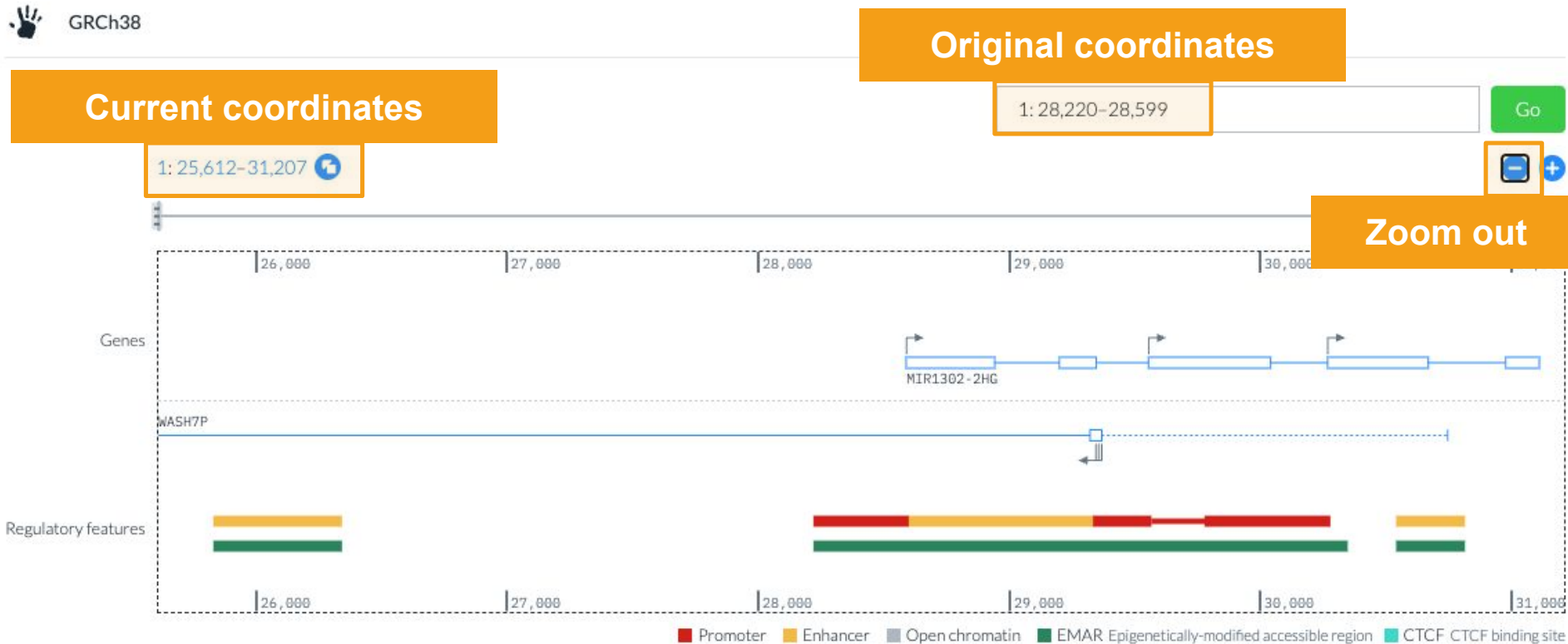
brain (120-122 dpf)

neural progenitor cell (l. 5 dpf)

SK-N-SH (l. 4 y)

H3K27ac H3K4me1 H3K27me2

Genome view tab: zoomed out



Genome view tab: chromosome change



GRCh38

Change chromosome

1: 28,220–28,599



Sequence ID

Search...

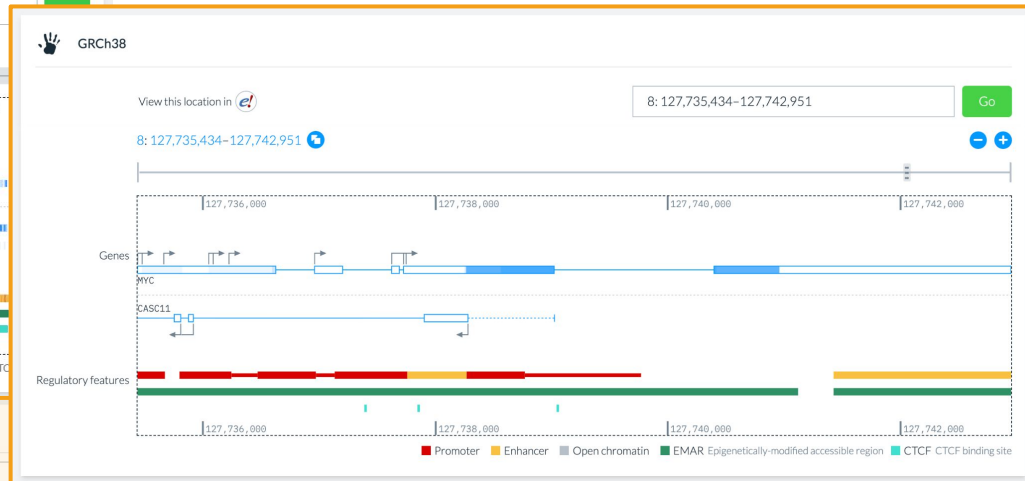
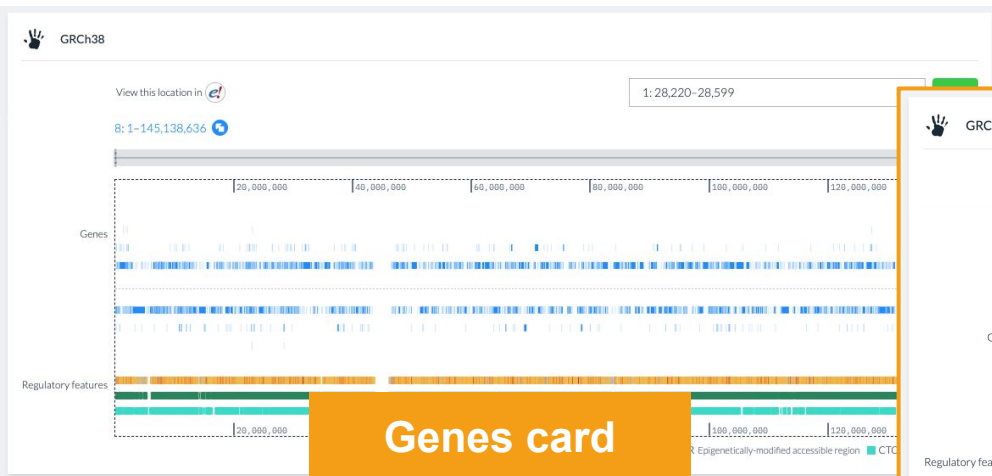
1	14
2	15
3	16
4	17
5	18
6	19
7	20
8	21
9	22

Regul

28,300

28,300

Genome view tab: Gene search



Genes 2305

Search:

15 entries per page

Showing 1 to 1 of 1 genes.

ID	Symbol	Biotype	Location
☐ ENSG00000136997.22 	MYC	protein_coding	8: 127,735,434-127,742,951

Navigation:   1  

Regulatory features 30668

Regulatory activity 144 epigenomes

Details can be viewed when the region is less than 2 Mbp.

Genome view tab: Gene selection



GRCh38

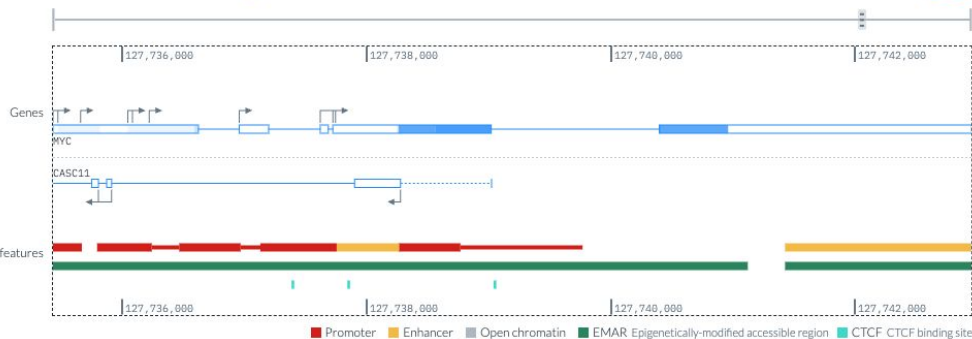
View this location in [EN](#)

8: 127,735,434-127,742,951

Go

8: 127,735,434-127,742,951

+



Genes 2



Search...

15 entries per page

Showing 1 to 2 of 2 genes.

ID	Symbol	Biotype	Location
☐ ENSG00000249375.9	CASC11	lncRNA	8: 127,686,343-127,739,022
☒ ENSG00000136997.22	MYC	protein_coding	8: 127,735,434-127,742,951

1

Gene selection



GRCh38

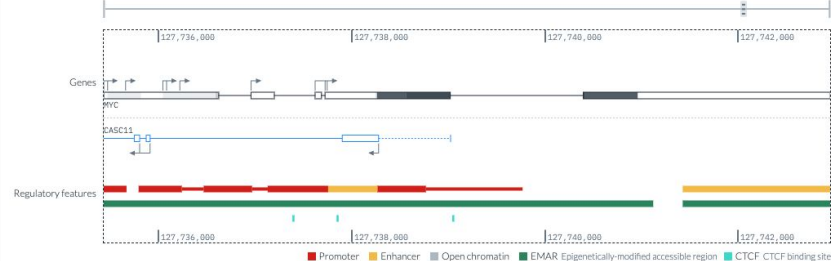
View this location in [EN](#)

8: 127,735,434-127,742,951

Go

8: 127,735,434-127,742,951

+



Genes 2

MYC(protein_coding) x



Search...

15 entries per page

Showing 1 to 2 of 2 genes.

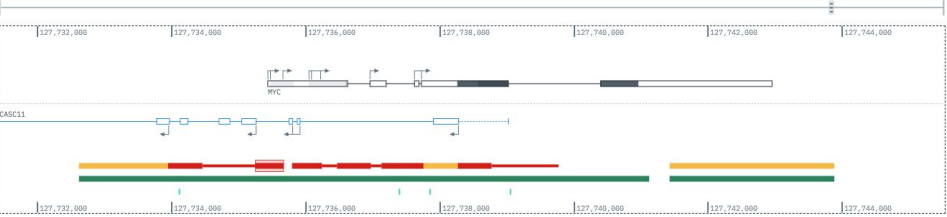
ID	Symbol	Biotype	Location
☐ ENSG00000249375.9	CASC11	lncRNA	8: 127,686,343-127,739,022
☒ ENSG00000136997.22	MYC	protein_coding	8: 127,735,434-127,742,951

1

Genome view tab: Regulatory feature selection

View this location in 

8:127,731,433-127,745,516  



Genes 2 **MYC (protein_coding)** 

Search...

15 entries per page Showing 1 to 2 of 2 genes.

ID	Symbol	Biotype	Location
☐ ENSG00000249375.9	CASC11	lncRNA	8:127,684,949-127,739,000
☒ ENSG00000136997.22	MYC	protein_coding	8:127,735,434-127,742,951

Regulatory features selection

Regulatory features 15 **ENSR8_93TNWH** 

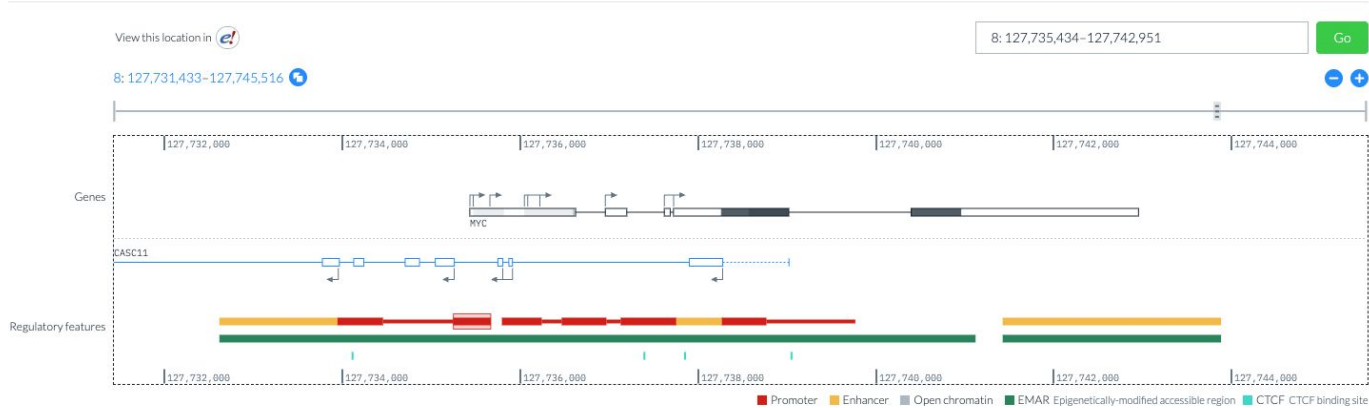
Search...

15 entries per page Showing 1 to 15 of 15 features.

ID	Type	Location	Genes
☐ ENSR8_CGN7Z	Enhancer	8:127,732,619-127,733,945	
☐ ENSR00001612796	EMAR	8:127,732,619-127,741,119	
☐ ENSR8_BXCG3	Promoter	8:127,733,946-127,734,457	CASC11 
☐ ENSR8_535379M	CTCF	8:127,734,105-127,734,126	
☒ ENSR8_93TNWH	Promoter	8:127,735,249-127,735,669	CASC11, MYC 
☐ ENSR8_93TNWM	Promoter	8:127,735,795-127,736,241	CASC11, MYC 
☐ ENSR8_93TNWT	Promoter	8:127,736,468-127,736,968	MYC 

Genome view tab: Regulatory activity

GRCh38



Genes 2 **MYC (protein_coding)** 

Regulatory features 15 **ENSR0_93TNWH** 

Regulatory activity

Regulatory activity 4 epigenomes

Epigenomes 

astrocyte

brain (120-122 dpf)

neural progenitor cell (f. 5 dpf)

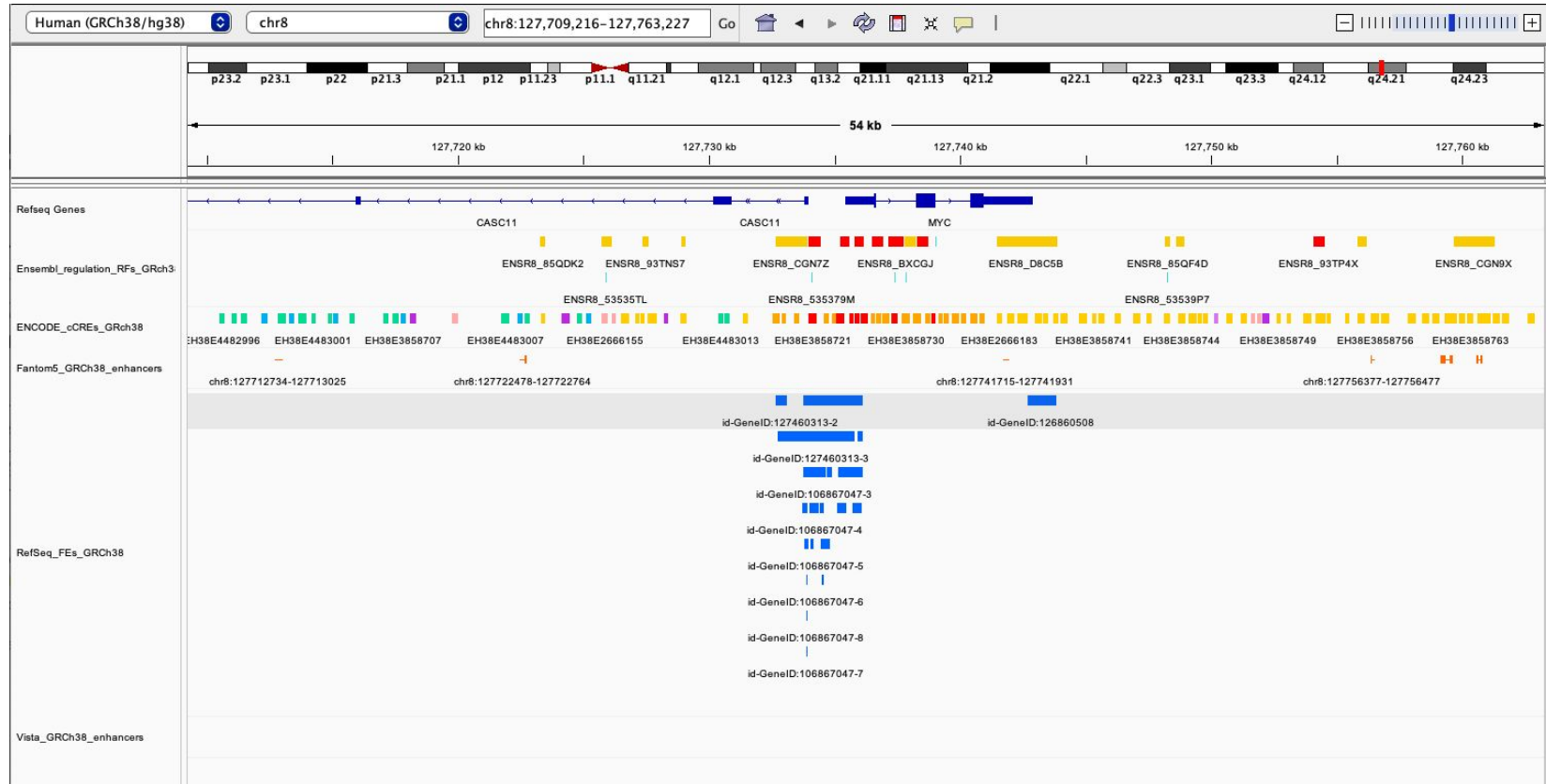
SK-N-SH (f. 4 y)

□ Open chromatin: Peak ■ Open chromatin: Signal ■ H3K4me3 ■ H3K27ac ■ H3K4me1 ■ H3K27me3

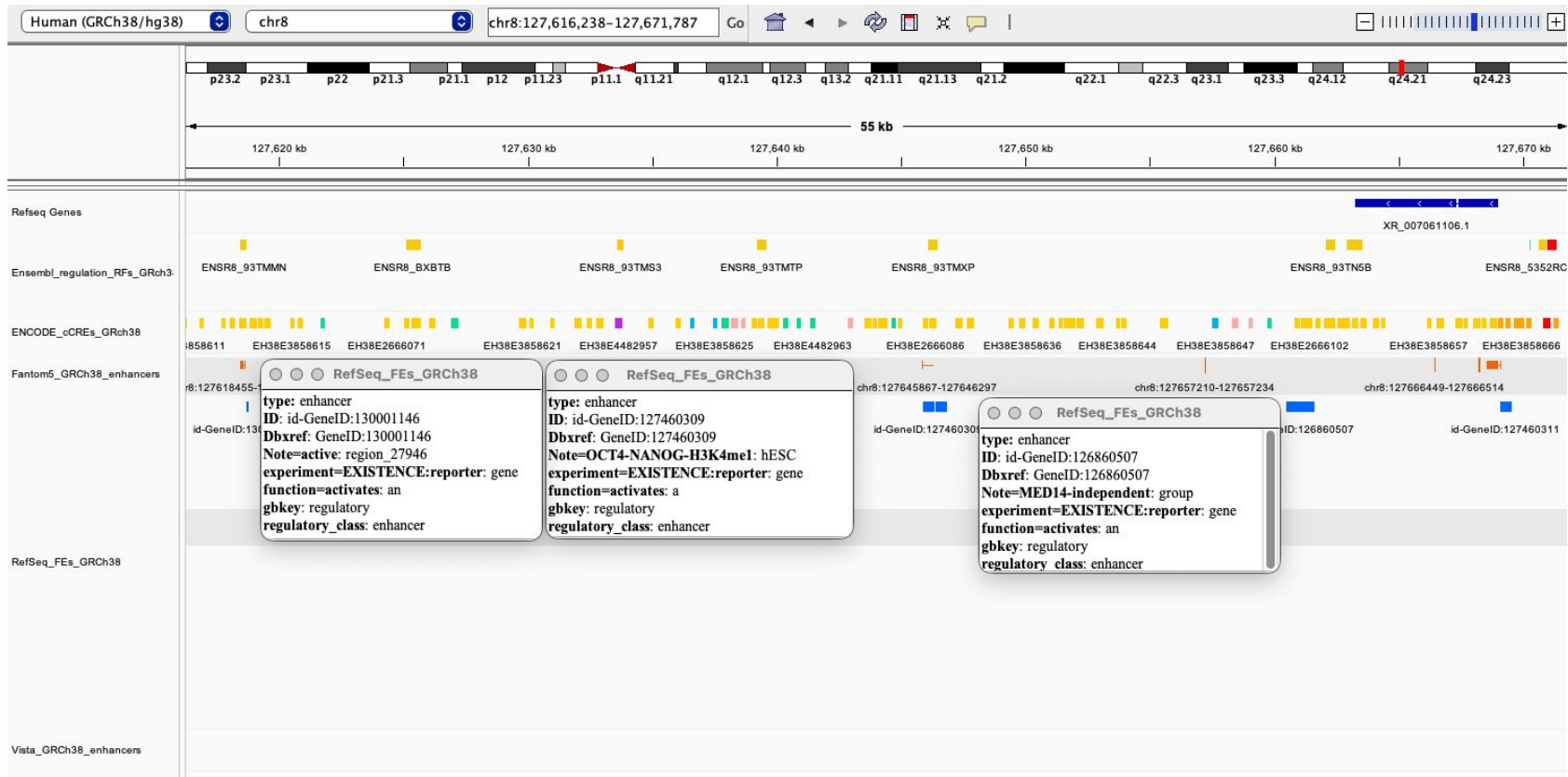
Differences between regulatory annotation frameworks

Resource	Main evidence categories	What it is useful for	Main caveat
<i>Ensembl Regulation</i>	Chromatin accessibility, histone modifications, CTCF ChIP-seq	Integrated regulatory features: promoters, enhancers, open chromatin regions, CTCF sites; activity across epigenomes	Evidence-based annotation, not direct proof of regulatory function
<i>ENCODE cCREs / SCREEN</i>	Chromatin accessibility, histone modifications, CTCF/TF ChIP-seq	Candidate cis-regulatory elements with promoter-like, enhancer-like, CTCF-like and other classes	Strong genome-wide catalogue, but still “candidate” elements
<i>RefSeq Functional Elements</i>	Manually curated experimental evidence, literature, protein binding, functional assays	High-confidence functional regions such as enhancers, silencers, insulators and locus control regions	Curated and evidence-rich, but less comprehensive genome-wide
<i>FANTOM5</i>	Transcription-based evidence, mainly CAGE	Active promoters and transcribed enhancers across many cell types/tissues	Best for active transcribed elements; may miss poised or non-transcribed CREs
<i>Roadmap Epigenomics / ChromHMM</i>	Chromatin accessibility, histone modifications	Chromatin-state annotations across many tissues/cell types	State labels are model-based and assay-dependent. Older technologies
<i>VISTA Browser</i>	Comparative genomics plus in vivo reporter assays	Experimentally validated developmental enhancers	Very valuable but focused on tested fragments, not comprehensive

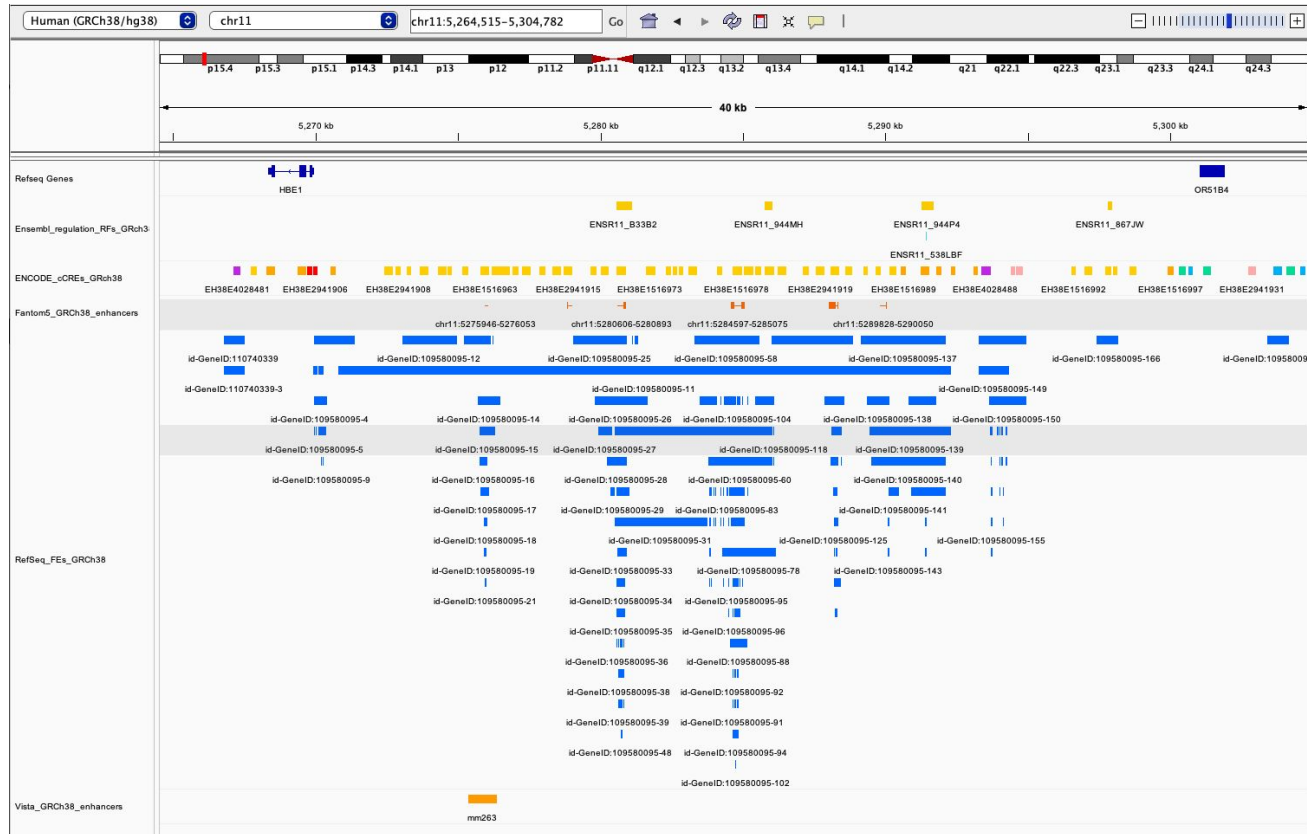
Differences between regulatory annotation frameworks: GRCh38



Differences between regulatory annotation frameworks: GRCh38



Differences between regulatory annotation frameworks: GRCh38



Hands-on



Ensembl Regulation

Answer the following questions:

- Explore the indicated locus on the Genome View tab.
- Display and explore regulatory features using the Regulation card.
- Filter epigenomes based on the given criteria
- Compare regulatory activity across the group of selected epigenomes
- Inspect supporting epigenomic evidence in the indicated epigenome
- Identify at least one context-dependent example.
- Compare the results with the ones obtained in the previous session.